1. Read, decode and make sense of statements... to create a model of the situation (Point 4)

- **Analysis:** This is the first step and the most core ability required to solve the problem. Students must read the text and the table, and understand that this is a question about **proportion**. They need to extract the logic from the information "100 mL recipe" and "60 mL salad oil" to build the correspondence between the total volume and the component volume.

2. Recognise mathematical structure (relationships, and patterns) (Point 5)

- **Analysis:** The problem asks to scale the recipe from 100 mL to 150 mL. Students need to identify the inherent **linear relationship** or **proportional structure**. That is: if the total volume increases by 1.5 times ($150 \div 100$), then all ingredients must increase by the same proportion.

3. Identify the key variables in a model (Point 2)

- **Analysis:** Amidst various data points (60mL oil, 30mL vinegar, 10mL soy sauce), students must identify the **key variables** needed to solve the specific question—namely, the "initial total (100mL)", the "target total (150mL)", and the "target component (60mL salad oil)". Ignoring the vinegar and soy sauce as irrelevant information is crucial for modeling.

4. Translate a problem into a standard mathematical representation or algorithm (Point 9)

- **Analysis:** Students need to translate the real-life scenario of ingredient ratios into a mathematical expression. For example, setting up the proportion $\frac{60}{100} = \frac{x}{150}$, or using the algorithm: $60 \times (150 \div 100)$.

5. Recognise aspects of a problem that correspond with known problems or mathematical concepts (Point 8)

- **Analysis:** Students need to associate this problem with concepts learned in class, such as "scaling factor" or "proportional distribution". Recognizing that this is essentially a **scaling** problem allows them to apply known multiplication and division procedures to solve it.

6. Identify and describe the mathematical aspects of a real-world problem... (Point 6)

- **Analysis:** The question transforms an everyday cooking scenario (making salad dressing) into a mathematical problem. Students need to identify the mathematical attributes within this real-world context (volume and ratio) and clarify that the "amount of oil" changes as the "total recipe amount" changes.

1. Read, decode and make sense of statements... to create a model of the situation (点 4)

- **分析：** 这是解决该题的第一步，也是最核心的能力。学生必须阅读文本和表格，理解这是一个关于**比例（Proportion）**的问题。需要从“100 mL 配方”和“60 mL 沙拉油”的信息中提取逻辑，构建出总量与分量之间的对应关系 (p. 1)。

2. Recognise mathematical structure (relationships, and patterns) (点 5)

- **分析：** 题目要求将配方从 100 mL 扩大到 150 mL。学生需要识别出其中的**线性关系**或**比例结构**。即：如果总量增加了 1.5 倍（$150 \div 100$），那么所有成分也必须按相同比例增加 (p. 1)。

3. Identify the key variables in a model (点 2)

- **分析：** 在众多的数据中（油 60mL、醋 30mL、酱油 10mL），学生必须识别出解决特定问题所需的**关键变量**——即“初始总量（100mL）”、“目标总量（150mL）”以及“目标成分（沙拉油 60mL）”。忽略醋和酱油等干扰信息是建模的关键 (p. 1)。

4. Translate a problem into a standard mathematical representation or algorithm (点 9)

- **分析：** 学生需要将现实生活中的调料配比转化为数学算式。例如，建立比例式 $\frac{60}{100} = \frac{x}{150}$，或者使用算法：$60 \times (150 \div 100)$ (p. 1)。

5. Recognise aspects of a problem that correspond with known problems or mathematical concepts (点 8)

- **分析：** 学生需要联想到课本中所学的“变化倍数”或“比例分配”等数学概念。识别出这本质上是一个**缩放（Scaling）**问题，并运用已知的乘除法程序来解决 (p. 1)。

6. Identify and describe the mathematical aspects of a real-world problem... (点 6)

- **分析：** 题目将一个日常生活的烹饪场景（做沙拉酱）转化为数学问题。学生需要识别出这一现实情境中的数学属性（体积和配比），并明确“油的用量”是随着“总配方量”变化而变化的自变量 (p. 1)。